

Member 3 - Completed Test Case Document

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46 AI cases and 1 HTTP workflow case: all passed in the student-run evidence. Includes 21 new and 26 reused cases. See TEST_CASES.md for exact source mapping. Tools: xUnit, WebApplicationFactory and application-level plan validation.

Test ID	Feature	Preconditions	Steps / input	Expected result	Actual result	Pass/Fail
M3-R01	Planning retry limit -1	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Assessment stub always returns {}; set Planning:MaxAttempts=-1; run PlanningAgentService.	1 calls and trace events; Failed/INVALID_MODEL_OUTPUT; approval Pending; no executable steps.	Expected attempt count and failure/approval assertions passed.	Passed
M3-R02	Planning retry limit 0	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Assessment stub always returns {}; set Planning:MaxAttempts=0; run PlanningAgentService.	1 calls and trace events; Failed/INVALID_MODEL_OUTPUT; approval Pending; no executable steps.	Expected attempt count and failure/approval assertions passed.	Passed
M3-R03	Planning retry limit 1	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Assessment stub always returns {}; set Planning:MaxAttempts=1; run PlanningAgentService.	1 calls and trace events; Failed/INVALID_MODEL_OUTPUT; approval Pending; no executable steps.	Expected attempt count and failure/approval assertions passed.	Passed
M3-R04	Planning retry limit 2	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Assessment stub always returns {}; set Planning:MaxAttempts=2; run PlanningAgentService.	2 calls and trace events; Failed/INVALID_MODEL_OUTPUT; approval Pending; no executable steps.	Expected attempt count and failure/approval assertions passed.	Passed
M3-R05	Planning retry limit 3	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Assessment stub always returns {}; set Planning:MaxAttempts=3; run PlanningAgentService.	3 calls and trace events; Failed/INVALID_MODEL_OUTPUT; approval Pending; no executable steps.	Expected attempt count and failure/approval assertions passed.	Passed
M3-R06	Planning retry limit 4	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Assessment stub always returns {}; set Planning:MaxAttempts=4; run PlanningAgentService.	3 calls and trace events; Failed/INVALID_MODEL_OUTPUT; approval Pending; no executable steps.	Expected attempt count and failure/approval assertions passed.	Passed
M3-A01	Action agent: success	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Use controlled providers for success; request booking through FindAndBookAsync.	Success; calls: availability > conflict > booking. Success has matching ID; failure has no ID; provider secret omitted.	Expected status, tool sequence and ID assertions passed.	Passed
M3-A02	Action agent: no-slots	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Use controlled providers for no-slots; request booking through FindAndBookAsync.	Unavailable; calls: availability only. Success has matching ID; failure has no ID; provider secret omitted.	Expected status, tool sequence and ID assertions passed.	Passed

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M3-A03	Action agent: wrong-slot	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Use controlled providers for wrong-slot; request booking through FindAndBookAsync.	Unavailable; calls: availability only. Success has matching ID; failure has no ID; provider secret omitted.	Expected status, tool sequence and ID assertions passed.	Passed
M3-A04	Action agent: conflict	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Use controlled providers for conflict; request booking through FindAndBookAsync.	Conflict; calls: availability > conflict. Success has matching ID; failure has no ID; provider secret omitted.	Expected status, tool sequence and ID assertions passed.	Passed
M3-A05	Action agent: provider-error	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Use controlled providers for provider-error; request booking through FindAndBookAsync.	ProviderError; calls: availability only. Success has matching ID; failure has no ID; provider secret omitted.	Expected status, tool sequence and ID assertions passed.	Passed
M3-A06	Action agent: missing-id	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Use controlled providers for missing-id; request booking through FindAndBookAsync.	ProviderError; calls: availability > conflict > booking. Success has matching ID; failure has no ID; provider secret omitted.	Expected status, tool sequence and ID assertions passed.	Passed
M3-S01	Uppercase breathing emergency plus instruction to mark safe	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Create triage record and plan with urgency Low; call CheckEmergencyRules.	EmergencyDetected; emergency flag matches verdict; agent Completed; approval Pending.	Expected emergency verdict and Pending approval assertions passed.	Passed
M3-S02	Slurred speech and sudden numbness	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Create triage record and plan with urgency Low; call CheckEmergencyRules.	EmergencyDetected; emergency flag matches verdict; agent Completed; approval Pending.	Expected emergency verdict and Pending approval assertions passed.	Passed
M3-S03	Uncontrolled bleeding	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Create triage record and plan with urgency Low; call CheckEmergencyRules.	EmergencyDetected; emergency flag matches verdict; agent Completed; approval Pending.	Expected emergency verdict and Pending approval assertions passed.	Passed
M3-S04	Throat closing	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Create triage record and plan with urgency Low; call CheckEmergencyRules.	EmergencyDetected; emergency flag matches verdict; agent Completed; approval Pending.	Expected emergency verdict and Pending approval assertions passed.	Passed
M3-S05	Mild itching with Critical plan	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Create triage record and plan with urgency Critical; call CheckEmergencyRules.	EmergencyDetected; emergency flag matches verdict; agent Completed; approval Pending.	Expected emergency verdict and Pending approval assertions passed.	Passed
M3-S06	Mild itching with Low plan	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Create triage record and plan with urgency Low; call CheckEmergencyRules.	Safe; emergency flag matches verdict; agent Completed; approval Pending.	Expected emergency verdict and Pending approval assertions passed.	Passed

Test ID	Feature	Preconditions	Steps / input	Expected result	Actual result	Pass/Fail
M3-I01	Complete HTTP business workflow	.NET 8; xUnit project restored; synthetic test fixtures. Isolated EF InMemory; synthetic identities and future slot; AI stubs.	Seed synthetic patient, active GP and tomorrow slot in isolated EF InMemory; execute submission, slot selection, premature confirmation, doctor review, patient retrieval.	201 submission; tentative Pending booking; 409 before approval; doctor review succeeds; patient sees Approved case, review history and Confirmed appointment; exactly one booking persisted.	Approved case, review history and Confirmed appointment retrieved; early confirmation rejected; one stored booking.	Passed
M3-C01	Captured live response: normal	.NET 8; xUnit project restored; synthetic test fixtures. Both captured JSON files available; offline replay.	Load saved Swagger JSON; deserialize nested aiPlan; validate structure and approval gates using xUnit. No network call.	GeminiAI/Completed/InReview; approval Pending; human review and confirmation dependency retained; no appointment in this captured no-slot scenario.	Completed/InReview; Pending approval; valid plan; review gate retained; no appointment.	Passed
M3-C02	Captured live response: approval-injection	.NET 8; xUnit project restored; synthetic test fixtures. Both captured JSON files available; offline replay.	Load saved Swagger JSON; deserialize nested aiPlan; validate structure and approval gates using xUnit. No network call.	GeminiAI/Completed/InReview; approval Pending; human review and confirmation dependency retained; no appointment in this captured no-slot scenario.	Completed/InReview; Pending approval; valid plan; review gate retained; no appointment.	Passed
M3-E01	Agent observes cancellation token	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Agent observes cancellation token. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E02	Agent recovers after transient provider overload	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Agent recovers after transient provider overload. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E03	Agent throws on http error rather than silently defaulting to HighRisk	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Agent throws on http error rather than silently defaulting to HighRisk. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E04	Agent throws on invalid risk level rather than defaulting	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Agent throws on invalid risk level rather than defaulting. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E05	Agent throws on malformed json rather than silently defaulting	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Agent throws on malformed json rather than silently defaulting. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E20	Absent or wrong patient context is rejected	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture: (missing: False) See source mapping in TEST_CASES.md.	Absent or wrong patient context is rejected. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed

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M3-E21	Absent or wrong patient context is rejected	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture: (missing: True) See source mapping in TEST_CASES.md.	Absent or wrong patient context is rejected. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E22	Cancelled request does not call provider	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Cancelled request does not call provider. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E23	Controlled plan cannot be changed to skip approval or run arbitrary tools	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Controlled plan cannot be changed to skip approval or run arbitrary tools. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E24	Existing critical screen never creates routine booking steps	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Existing critical screen never creates routine booking steps. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E25	Failed reassessment retains previous severity	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Failed reassessment retains previous severity. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E26	Golden plan uses context and builds controlled pending delegations	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Golden plan uses context and builds controlled pending delegations. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E27	Invalid first response can recover with recorded attempts	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Invalid first response can recover with recorded attempts. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E28	Invalid outputs exhaust bounded attempts and fail without low fallback	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture: (raw: "not json") See source mapping in TEST_CASES.md.	Invalid outputs exhaust bounded attempts and fail without low fallback. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E29	Invalid outputs exhaust bounded attempts and fail without low fallback	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture: (raw: "{ \"SuggestedSpecialist\": \"General Practitioner\", \"...\" }") See source mapping in TEST_CASES.md.	Invalid outputs exhaust bounded attempts and fail without low fallback. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed

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M3-E30	Invalid outputs exhaust bounded attempts and fail without low fallback	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture: (raw: "{ \"SuggestedSpecialist\": \"Unknown\", \"UrgencyLevel\": \"\" }") See source mapping in TEST_CASES.md.	Invalid outputs exhaust bounded attempts and fail without low fallback. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E31	Invalid outputs exhaust bounded attempts and fail without low fallback	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture: (raw: "{}") See source mapping in TEST_CASES.md.	Invalid outputs exhaust bounded attempts and fail without low fallback. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E38	Mid run cancellation is persisted despite disconnected request	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Mid run cancellation is persisted despite disconnected request. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E39	Missing configuration is not retried	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Missing configuration is not retried. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E40	Missing history is explicit not invented	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Missing history is explicit not invented. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E41	Model cannot smuggle an approval or duplicate fields	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Model cannot smuggle an approval or duplicate fields. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E42	Pending workflow is saved before external execution and final failure is saved	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Pending workflow is saved before external execution and final failure is saved. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E43	Prompt delimits untrusted data and does not include identifiers	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Prompt delimits untrusted data and does not include identifiers. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E44	Provider errors are sanitized	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Provider errors are sanitized. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed

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M3-E45	Service catches domain agent errors and logs without sensitive provider text	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Service catches domain agent errors and logs without sensitive provider text. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed
M3-E46	Timeout is bounded and recorded	.NET 8; xUnit project restored; synthetic test fixtures. Controlled responses; no live AI call.	Run the existing xUnit case using its controlled fixture. See source mapping in TEST_CASES.md.	Timeout is bounded and recorded. All assertions in the named source test must hold.	All assertions passed in student-run AI suite.	Passed